

BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 3 Practical No: 1

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Subject Name : Deep Learning

Problem Statement :

To install and configure TensorFlow/Keras in Google Colab and perform data preprocessing, normalization, train-test split, and visualization on a sample dataset. The objective is to prepare the dataset for machine learning tasks and verify the TensorFlow/Keras environment.

Algorithm :

1. Open Google Colab and create a new notebook.
2. Install TensorFlow (if required) and verify its installation.
3. Import the necessary libraries such as TensorFlow, Keras, NumPy, Pandas, Matplotlib, and Scikit-learn.
4. Load a sample dataset (Iris dataset).
5. Convert the dataset into a Pandas DataFrame.
6. Explore the dataset by displaying its structure, summary statistics, and checking for missing values.
7. Separate the features (X) and target variable (y).
8. Normalize the feature values using StandardScaler.

9. Split the dataset into training and testing sets using the train-test split technique.
10. Visualize the dataset using scatter plots, histograms, and box plots.
11. Create a simple Keras Sequential model to verify that TensorFlow/Keras has been configured successfully.
12. Execute the notebook and observe the outputs.

Code / Output:

```
[2]
✓ 4s
import tensorflow as tf

print("TensorFlow Version:", tf.__version__)
```

TensorFlow Version: 2.20.0

```
[3]
✓ 2s
import tensorflow as tf
from tensorflow import keras

import numpy as np
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
```

```
[4]
✓ 0s
iris = load_iris()

X = iris.data
y = iris.target
```

```
[5]
✓ 0s
df = pd.DataFrame(X, columns=iris.feature_names)
df["Target"] = y

df.head()
```

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	Target
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

Next steps: [Generate code with df](#) [New interactive sheet](#)

[6]
✓ 0s df.info()

... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
Column Non-Null Count Dtype

0 sepal length (cm) 150 non-null float64
1 sepal width (cm) 150 non-null float64
2 petal length (cm) 150 non-null float64
3 petal width (cm) 150 non-null float64
4 Target 150 non-null int64
dtypes: float64(4), int64(1)
memory usage: 6.0 KB

[7]
✓ 0s df.describe()

...

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	Target
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333	1.000000
std	0.828066	0.435866	1.765298	0.762238	0.819232
min	4.300000	2.000000	1.000000	0.100000	0.000000
25%	5.100000	2.800000	1.600000	0.300000	0.000000
50%	5.800000	3.000000	4.350000	1.300000	1.000000
75%	6.400000	3.300000	5.100000	1.800000	2.000000
max	7.900000	4.400000	6.900000	2.500000	2.000000

[8]
✓ 0s df.isnull().sum()

0
sepal length (cm) 0
sepal width (cm) 0
petal length (cm) 0
petal width (cm) 0
Target 0
dtype: int64

[9]
✓ 0s X = df.iloc[:, :-1]
y = df["Target"]

[10]
✓ 0s scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

[11]
✓ Os

```
x_train, x_test, y_train, y_test = train_test_split(  
    x_scaled,  
    y,  
    test_size=0.2,  
    random_state=42  
)
```

[12]
✓ Os

```
print(x_train.shape)  
print(x_test.shape)
```

▼

```
(120, 4)  
(30, 4)
```

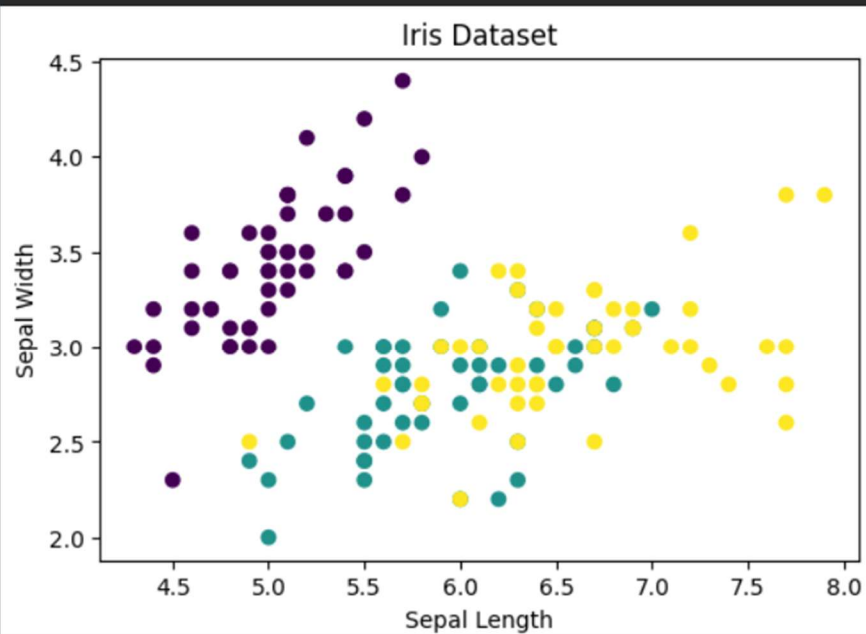
[13]
✓ Os



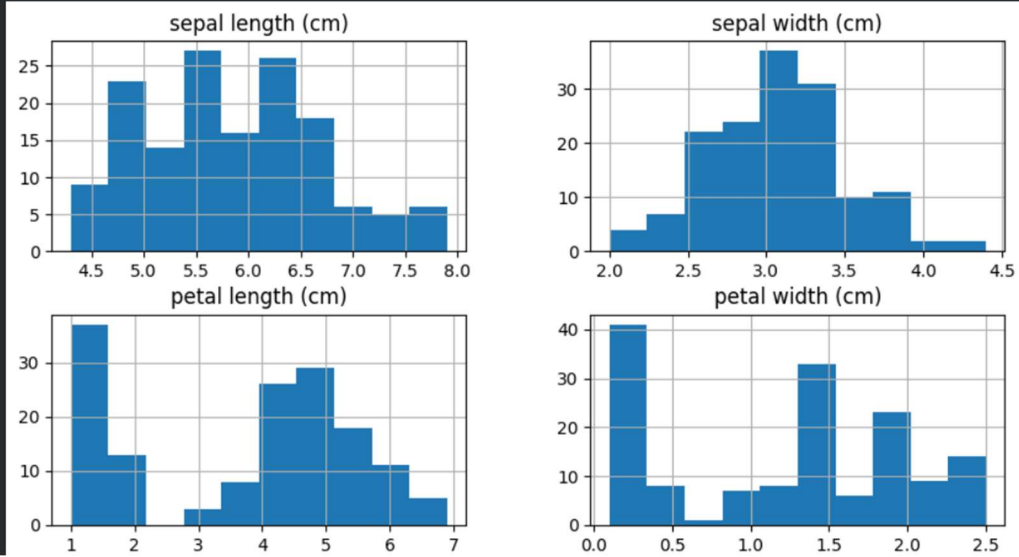
```
plt.figure(figsize=(6,4))  
plt.scatter(df.iloc[:,0], df.iloc[:,1], c=y)  
plt.xlabel("Sepal Length")  
plt.ylabel("Sepal Width")  
plt.title("Iris Dataset")  
plt.show()
```

▼

...



```
[14] df.hist(figsize=(10,8))  
plt.show()
```



```
[15] df.boxplot(figsize=(10,5))  
plt.show()
```

